2.38 Suppose that the formula ϕ is constructed by taking subformulas $\theta_1, \theta_2, ..., \theta_n$ in that order and joining them together only using the connective $\wedge$, with brackets inserted in such a way as to make ϕ a formula. Show that ϕ is true precisely for those truth assignments which make each of the subformulas $\theta_1, \theta_2, ..., \theta_n$ true. Deduce that any two such formulas (using the same $\theta_1, \theta_2, ..., \theta_n$ in that order) are logically equivalent. [*Hint:* Use the version of mathematical induction with the hypothesis that the result holds for all $k \leq n$ where $n \geq 1$.]

Proof. We proceed by induction on the number of θ's available in ϕ. ϕ is constructed by taking subformulas $\theta_1, \theta_2, ..., \theta_n$ in that order and joining them together only using the connective $\wedge$, with brackets inserted in such a way as to make ϕ a formula. We will show ϕ is true precisely for those truth assignments which make each of the subformulas $\theta_1, \theta_2, ..., \theta_n$ true.

Base Case: Suppose that $\phi = \theta_1$. Let v be any truth assignment. Therefore $v(\phi) = T \iff v(\theta_1) = T$.

Inductive Hypothesis: Let $n \geq 1$. Suppose that for every $k \leq n$ the following holds: if ψ is a formula built from the formulas $\psi_1, \psi_2, \ldots, \psi_k$ in that order, joined only by the connective $\wedge$ with brackets inserted in such a way as to make ψ a formula, then for every truth assignment v,

$$v(\psi) = T \iff v(\psi_j) = T \text{ for all } j \text{ with } 1 \leq j \leq k.$$

Inductive Step: Let ϕ be built from the formulas $\theta_1, \theta_2, ..., \theta_n, \theta_{n+1}$ in that order, joined only by the connective $\wedge$ with brackets inserted in such a way as to make ϕ a formula.

ϕ has a principal connective, so let θ_i be immediately to the left of the principal connective and let θ_{i+1} be immediately to the right of the principal connective where $1 \leq i \leq n$.

Therefore ϕ can be rewritten as $\phi = \alpha \wedge \beta$ where α is built from the formulas $\theta_1, \theta_2, ..., \theta_i$ and β is built from the formulas $\theta_{i+1}, ..., \theta_n, \theta_{n+1}$ in that order, joined only by the connective $\wedge$ with brackets inserted in such a way as to make them each valid subformulas of ϕ.yea

Since $i \leq n$ and $(n+1) - i \leq n$ the inductive hypothesis applies to α, β.

Let v be any truth assignment. Since $\phi = \alpha \wedge \beta$, the truth table for $\wedge$ gives

$$v(\phi) = T \iff v(\alpha) = T = v(\beta).$$

By the inductive hypothesis with $k = i$ and $\psi_1, \ldots, \psi_i := \theta_1, \ldots, \theta_i$,

$$v(\alpha) = T \iff v(\theta_j) = T \text{ for all } j \text{ with } 1 \leq j \leq i.$$

By the inductive hypothesis with $k = (n+1) - i$ and $\psi_1, \ldots, \psi_{(n+1)-i} := \theta_{i+1}, \ldots, \theta_{n+1}$,

$$v(\beta) = T \iff v(\theta_j) = T \text{ for all } j \text{ with } i+1 \leq j \leq n+1.$$

Combining these three biconditionals,

$$v(\phi) = T \iff v(\theta_j) = T \text{ for all } j \text{ with } 1 \leq j \leq n+1,$$

which is precisely the claim for $n+1$. This completes the induction. $\square$

Deduction: Suppose ϕ and ϕ' are both built from the formulas $\theta_1, \theta_2, \ldots, \theta_n$ in that order, joined only by the connective $\wedge$ with brackets inserted in such a way as to make each a formula. Let v be any truth assignment. By the result above applied to each formula,

$$v(\phi) = T \iff v(\theta_j) = T \text{ for all } j \text{ with } 1 \leq j \leq n \iff v(\phi') = T.$$

If $v(\phi) = T$, then $v(\phi') = T$. If $v(\phi) = F$, then $v(\phi) \neq T$, so $v(\phi') \neq T$, and by bivalence $v(\phi') = F$. In either case $v(\phi) = v(\phi')$. Since v was arbitrary, $\phi \equiv \phi'$.